

16x8 FIFO Memory using Verilog RTL

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Simulation Waveform Explanation

1. Reset Operation

The reset signal is initially HIGH, which initializes the FIFO.

- Write pointer = 0
- Read pointer = 0
- Count = 0
- Empty = 1
- Full = 0

2. Write Operation

After reset is deasserted, wr_en becomes HIGH and four 8-bit values are written:

0x11, 0x22, 0x33, 0x44.

The write pointer increments from 0 to 4 and count increases from 0 to 4.

The empty flag becomes 0 while full remains 0.

3. Read Operation

The rd_en signal is asserted and data is read in FIFO order:

0x11, 0x22, 0x33, 0x44.

The read pointer increments from 0 to 4 while count decreases back to 0.

4. Empty Flag

After the last read, count becomes 0 and empty becomes 1, indicating the FIFO is empty.

5. Pointer Verification

Both write and read pointers increment correctly and end at the same location after all data is read.

Observations

- Reset initializes the FIFO correctly.
- Data is written successfully.
- Data is read in the same order it was written.
- Pointers increment correctly.
- Counter tracks stored elements correctly.
- Empty flag works correctly.
- Full flag remains LOW because only four of sixteen locations are used.

Conclusion

The waveform verifies that the 16x8 FIFO memory operates correctly. The FIFO preserves First-In-First-Out ordering, performs synchronous read/write operations, updates pointers and count correctly, and generates accurate full and empty status signals.